Wanyu Lin

Affiliation Department of Computing *Email:* wanylin@comp.polyu.edu.hk

Hong Kong Polytechnic University
PQ713a, Mong Man Wai Building
Hung Hom, Kowloon, Hong Kong

Personal Information *Citizenship* Chinese

Research Interests Interpretability of Graph Neural Networks, Social Media Analytics, Graph-Based Semi-Supervised Learning, Machine Learning in Adversarial Settings

Education

University of Toronto, Toronto, Ontario, Canada
Department of Electrical and Computer Engineering

- ◇ **Ph.D.**, Electrical and Computer Engineering, 2015 – 2020
  - ▷ *Dissertation:* "Social Media Analytics with Graph Convolutional Networks"
  - ▷ *Advisor:* Baochun Li, Department of Electrical and Computer Engineering

Hong Kong Polytechnic University, Hong Kong, China
Department of Computing

- ◇ **M.Phil.**, Computer Science, 2012 – 2015
  - ▷ *Dissertation:* "Delay-Constrained and Energy-Efficient Distributed Algorithms for Computation-Intensive Applications in WSNs"
  - ▷ *Advisor:* Jiannong Cao, Department of Computing

Huazhong University of Science and Technology, Wuhan, China
School of Electronic Information and Communications

- ◇ **B.Engr.**, Telecommunication Engineering, 2008 – 2012

Research Experience

Research Assistant, supervised by Prof. Baochun Li **2015 – 2020**
University of Toronto, Toronto

Research Assistant, supervised by Prof. Jiannong Cao **2012 – 2015**
Hong Kong Polytechnic University, Hong Kong

Publications

Wanyu Lin, Hao Lan, Baochun Li, "Generative Casual Explanations for Graph Neural Networks," in the Proceedings of the 38th International Conference on Machine Learning (ICML), July 18-24, 2021 (acceptance ratio: 21.4%)

Wanyu Lin, Baochun Li, "*Medley:* Predicting Social Trust in Time-Varying Online Social Networks," in the Proceedings of IEEE International Conference on Computer Communications (IN-

FOCOM), May 10-13, 2021 (acceptance ratio: 19.9%).

Wanyu Lin, Helei Cui, Baochun Li, Cong Wang, "Secure Similarity Search with Efficient Updates for Distributed High-Dimensional Datasets," in IEEE Transactions on Parallel and Distributed Systems (TPDS).

Wanyu Lin, Zhaolin Gao [1], Baochun Li, "*Guardian*: Evaluating Trust in Online Social Networks with Graph Convolutional Networks," in the Proceedings of IEEE International Conference on Computer Communications (INFOCOM), Toronto, Canada, July 6-9, 2020 (acceptance ratio: 20%).

Wanyu Lin, Zhaolin Gao, Baochun Li, "Shoestring: Graph-Based Semi-Supervised Learning with Severely Limited Data," in the Proceedings of the 2020 IEEE Conference on Computer Vision and Pattern Recognition (CVPR 2020), Seattle, Washington, June 16-18, 2020 (acceptance ratio: 22%).

Wanyu Lin, Shengxiang Ji [2], Baochun Li, "Adversarial Attacks on Link Prediction Algorithms Based on Graph Neural Networks," in the Proceedings of the 15th ACM ASIA Conference on Computer and Communications Security (ACM ASIACCS 2020), Taipei, Taiwan, June 1-5, 2020 (acceptance ratio: 21.7%).

Wanyu Lin, Jiannong Cao, Xuefeng Liu, "Distributed Least Squares Estimation Algorithms Design for Sensor Networks," Ad Hoc & Sensor Wireless Networks, Vol. 43 Issue 1/2, p85-108. 24p, 2019.

Wanyu Lin, Xu Yuan, Baochun Li, Cong Wang, "Multi-Client Searchable Encryption over Distributed Key-Value Store," in the Proceedings of the IEEE International Conference on Smart Computing (SMARTCOMP 2017), Hong Kong, China, May 29-31, 2017.

Wanyu Lin, Jiannong Cao, Xuefeng Liu, "E^3: Towards Energy-Efficient Distributed Least Squares Estimation in Sensor Networks," in the Proceedings of theIEEE/ACM International Symposium on Quality of Service (IWQoS 2014), Hong Kong, China, May 26-27, 2014.

Submitted Papers

Wanyu Lin, Baochun Li, "Status-Aware Heterogeneous Signed Network Embedding," submitted to TKDE 2021.

Wanyu Lin, Baochun Li, Cong Wang, "Efficient Multi-Keyword Fuzzy Search in Encrypted Distributed Key-Value Stores," submitted to IEEE Transactions on Parallel and Distributed Systems (TPDS), 2020.

Scholarly Talks

***Medley*: Predicting Social Trust in Time-Varying Online Social Networks** **May 2021**
IEEE INFOCOM, Virtual Conference

Adversarial Attacks on Link Prediction Algorithms Based on Graph Neural Networks October 2020

[1] First-year undergraduate research intern
[2] Second-year undergraduate research intern

ACM AsiaCCS, Virtual Conference

***Guardian*: Evaluating Trust in Online Social Networks with Graph Convolutional Networks July 2020**
IEEE INFOCOM, Virtual Conference

Shoestring: Graph-Based Semi-Supervised Learning with Severely Limited Data June 2020
IEEE CVPR, Virtual Conference

Distributed Least Squares Estimation Algorithms Design for Sensor Networks 2015
Wuhan University, Wuhan, China

Towards Energy-Efficient Distributed Least Squares Estimation in Sensor Networks May 2014
IEEE IWQoS, Hong Kong, China

Professional Services

- ◇ Technical Program Committee: IEEE SmartMM 2017
- ◇ Conference Helper: ACM CCS 2018, IEEE ICNP 2017, IEEE IWQoS 2014
- ◇ Reviewer for Journal Manuscript Submissions: IEEE TIFS, IEEE TKDE, IEEE TNSE, IEEE Access
- ◇ Reviewer for Conference Manuscript Submissions: ICML, NeurIPS, ACM Multimedia, USENIX HotCloud, ACM NOSSDAV, ACM MMSys, ACM/IEEE IoTDI, IFIP Networking, IEEE IWQoS, IEEE IC2E, WiOpt

Honors and Awards

- ◇ INFOCOM Student Conference Award, IEEE Communications Society, 2020.
- ◇ SGS Conference Grant, School of Graduate Studies, University of Toronto, 2020.
- ◇ Doctoral Completion Award, Department of ECE, University of Toronto, 2019 - 2020.
- ◇ University of Toronto Fellowship, Department of ECE, University of Toronto, 2015 - 2019.
- ◇ Edward S. Rogers Sr. Graduate Scholarships, University of Toronto, 2015 - 2019.
- ◇ Master of Philosophy Studentship, Hong Kong Polytechnic University, 2012 - 2015.
- ◇ Guan Zhigang Scholarship, Huazhong University of Science and Technology, Hong Kong Alumni Association, 2014.
- ◇ Outstanding Graduates Award, Huazhong University of Science and Technology, 2012.
- ◇ National Encouragement Scholarship, Huazhong University of Science and Technology, 2010 & 2011.
- ◇ Outstanding Individual Scholarship, Huazhong University of Science and Technology, 2010 & 2011.
- ◇ Outstanding Student Scholarship, Huazhong University of Science and Technology, 2010 &

2011.

◊ Self-Renewal Scholarship, Huazhong University of Science and Technology, 2008 & 2009.

Teaching Experience

Instructor Assistant **Sept. 2020 – Present**
University of Toronto, Toronto
Conduct online TA training sessions and assist the instructor in online course management.

Teaching Assistant **2016 – Present**
University of Toronto, Toronto

- ▷ ECE 421: Introduction to Machine Learning **Fall 2020**
- ▷ ECE 244: Programming Fundamentals (C++) **Fall 2017, Fall 2018, Fall 2019, Fall 2020**
- ▷ APS 105: Computer Fundamentals (C) **Winter 2018, Winter 2019, Summer & Winter 2020**
- ▷ ECE 110: Electrical Fundamentals **Winter 2016, Summer 2016, Winter 2017**
- ▷ ECE 341: Dynamic Systems and Control **Fall 2016**

Mentoring

Hao Zhang, undergraduate student at *University of Toronto* **Sept. 2019 – Apr. 2020**
Thesis title:"Applications of Convolutional Networks in the Context of Social Networks"
Anas Akram Musharraf, undergraduate student at *University of Toronto* **Sept. 2019 – Apr. 2020**
Thesis title:"Explaining Graph Neural Networks"

References

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Jiannong Cao, IEEE Fellow, Chair Professor
Distinguished Member of ACM
Senior Member of CCF
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